

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

3. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

3. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

css

mactime -b body.txt > timeline.txt

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

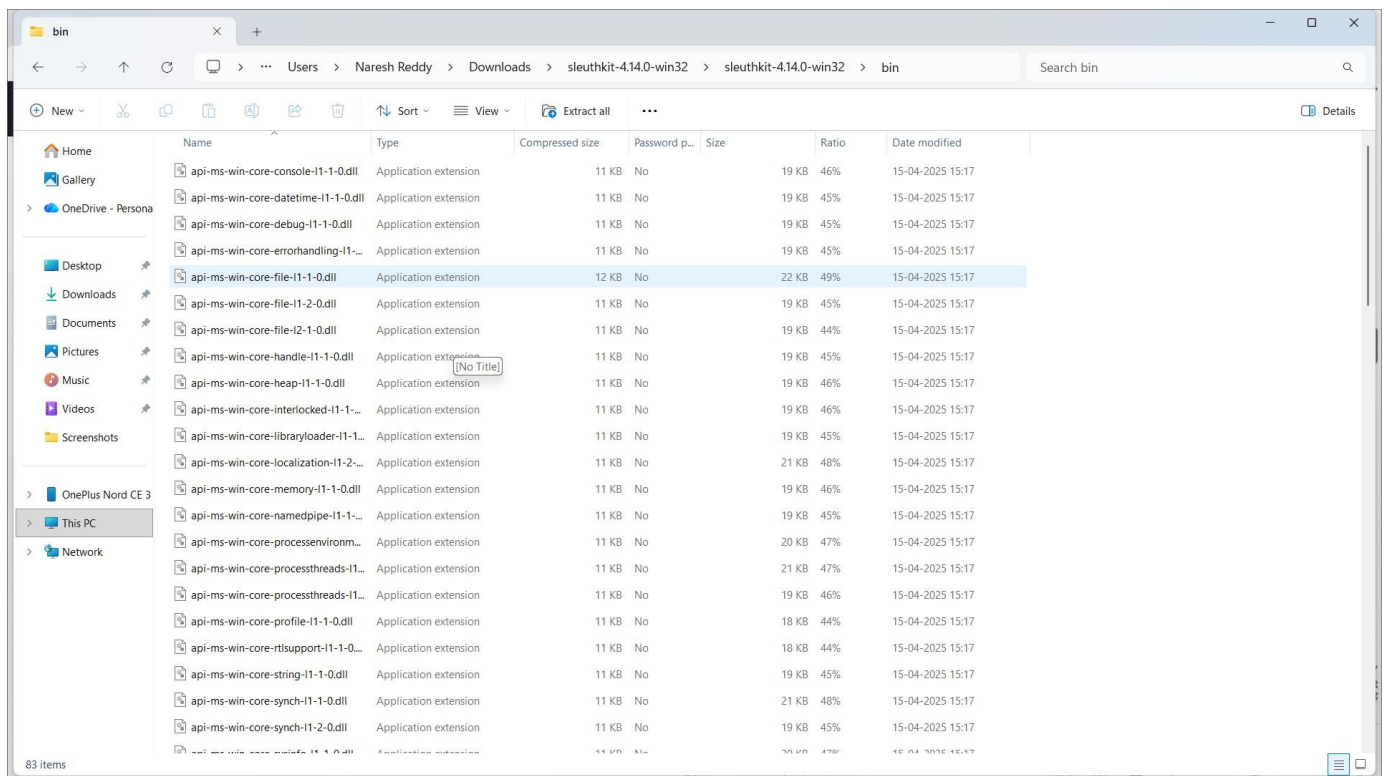
- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Fsstat command execution:



Identifying File System:

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Naresh Reddy>cd "C:\Users\Naresh Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin"

C:\Users\Naresh Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat
Missing image name
usage: fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
  -t: display type only
  -i imgtype: The format of the image file (use '-i list' for supported types)
  -b dev_sector_size: The size (in bytes) of the device sectors
  -f fstype: File system type (use '-f list' for supported types)
  -o imgoffset: The offset of the file system in the image (in sectors)
  -P pooltype: Pool container type (use '-P list' for supported types)
  -B pool_volume_block: Starting block (for pool volumes only)
  -v: verbose output to stderr
  -V: Print version
  -k password: Encryption password for encrypted volumes

C:\Users\Naresh Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>|
```

Installed Sleuth Kit:

```
C:\WINDOWS\system32\cmd. x + v
C:\Users\Naresh Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -i ewf -o 63 "C:\Users\Naresh Reddy\Downloads\4Dell Latitude CPi.E0
1"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP
METADATA INFORMATION
-----
First Cluster of MFT: 2897152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5
CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415
$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
C:\Users\Naresh Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

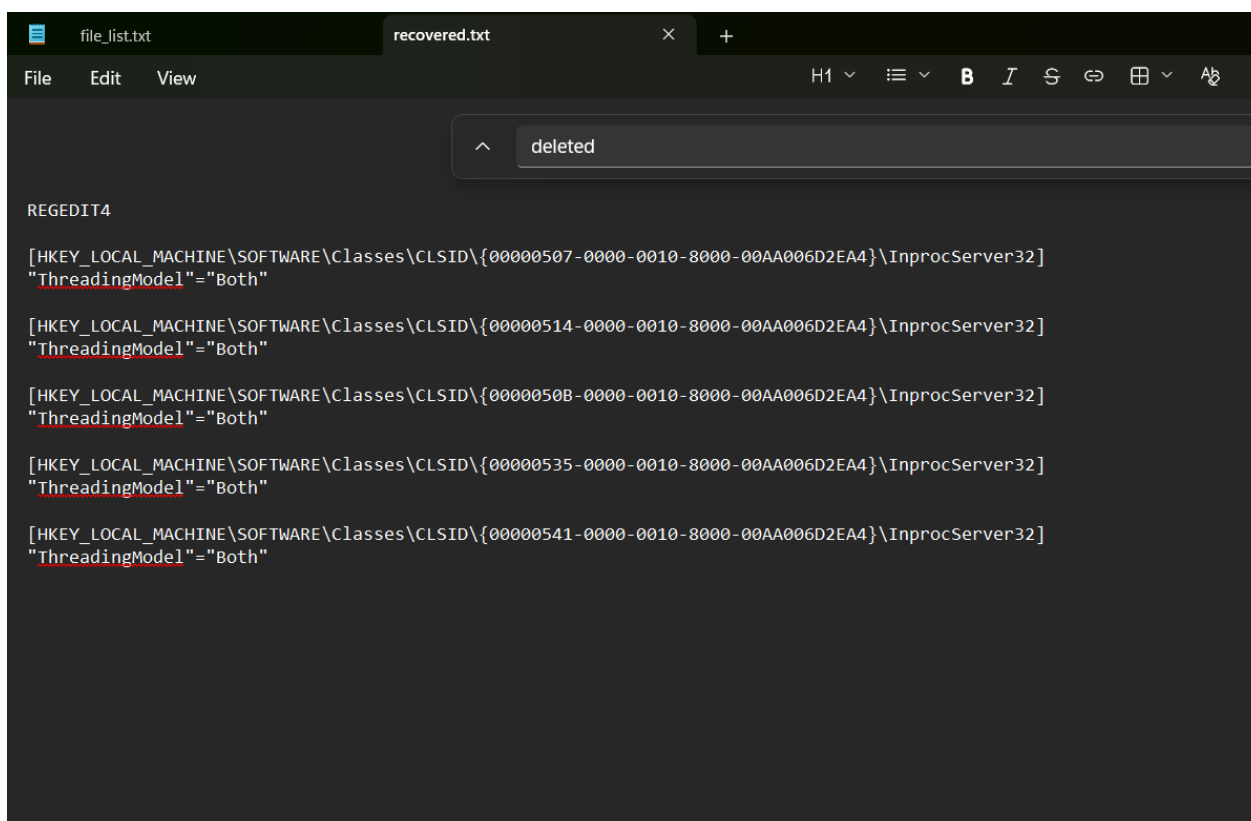
Identifying File System:

```
C:\Users\Naresh Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf -o 63 -r "C:\Users\Naresh Reddy\Downloads\4Dell Latitude CPi.E0
1" > file_lists.txt
```

```
Campus Creative Club Bank Details.xlsx file_lists.txt
File Edit View H1 I B I S S T V A
d/d 3671-144-7: Documents and Settings
+ d/d 3673-144-7: All Users
++ d/d 3707-144-7: Application Data
+++ r/r 3800-128-1: desktop.ini
+++ d/d 3708-144-7: Microsoft
++++ d/d 3709-144-1: Crypto
+++++ d/d 3712-144-1: DSS
+++++ d/d 3713-144-1: MachineKeys
+++++ d/d 3710-144-1: RSA
+++++ d/d 3711-144-1: MachineKeys
++++ d/d 10977-144-1: Dr Watson
+++++ r/r 10979-128-3: drwtsn32.log
+++++ r/r 10980-128-3: user.dmp
+++ d/d 7325-144-1: HTML Help
+++ d/d 7287-144-7: Media Index
+++++ r/r 6074-128-5: wmplibrary_v_0_12.db
+++ d/d 4759-144-1: Network
+++++ d/d 4762-144-1: Connections
+++++ d/d 4763-144-1: Pbk
+++++ r/r 10982-128-1: rasphone.pbk
+++++ r/r 4765-128-4: sharedaccess.ini
+++ d/d 5663-144-7: User Account Pictures
+++++ d/d 5664-144-7: Default Pictures
+++++ r/r 5665-128-4: airplane.bmp
+++++ r/r 5666-128-4: astronaut.bmp
+++++ r/r 5667-128-4: ball.bmp
+++++ r/r 5674-128-4: beach.bmp
+++++ r/r 5668-128-4: butterfly.bmp
+++++ r/r 5675-128-4: car.bmp
+++++ r/r 5669-128-4: cat.bmp
+++++ r/r 5676-128-4: chess.bmp
+++++ r/r 5677-128-4: dirt bike.bmp
+++++ r/r 5678-128-4: dog.bmp
+++++ r/r 5679-128-4: drip.bmp
+++++ r/r 5680-128-4: duck.bmp
+++++ r/r 5670-128-4: fish.bmp
+++++ r/r 5681-128-4: frog.bmp
+++++ r/r 5672-128-4: guitar.bmp
+++++ r/r 5682-128-4: horses.bmp
+++++ r/r 5683-128-4: kick.bmp
+++++ r/r 5684-128-4: lift-off.bmp
+++++ r/r 5685-128-4: palm tree.bmp
+++++ r/r 5671-128-4: pink flower.hmn
Ln 1, Col 1 414,033 characters Plain text 100% Windows (CRLF) UTF-8
```

Recovering Deleted File:

```
C:\Users\kmadh\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -i ewf -o 63 "C:\Users\kmadh\Downloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01" 150 > recovered.txt
```



Metadata:

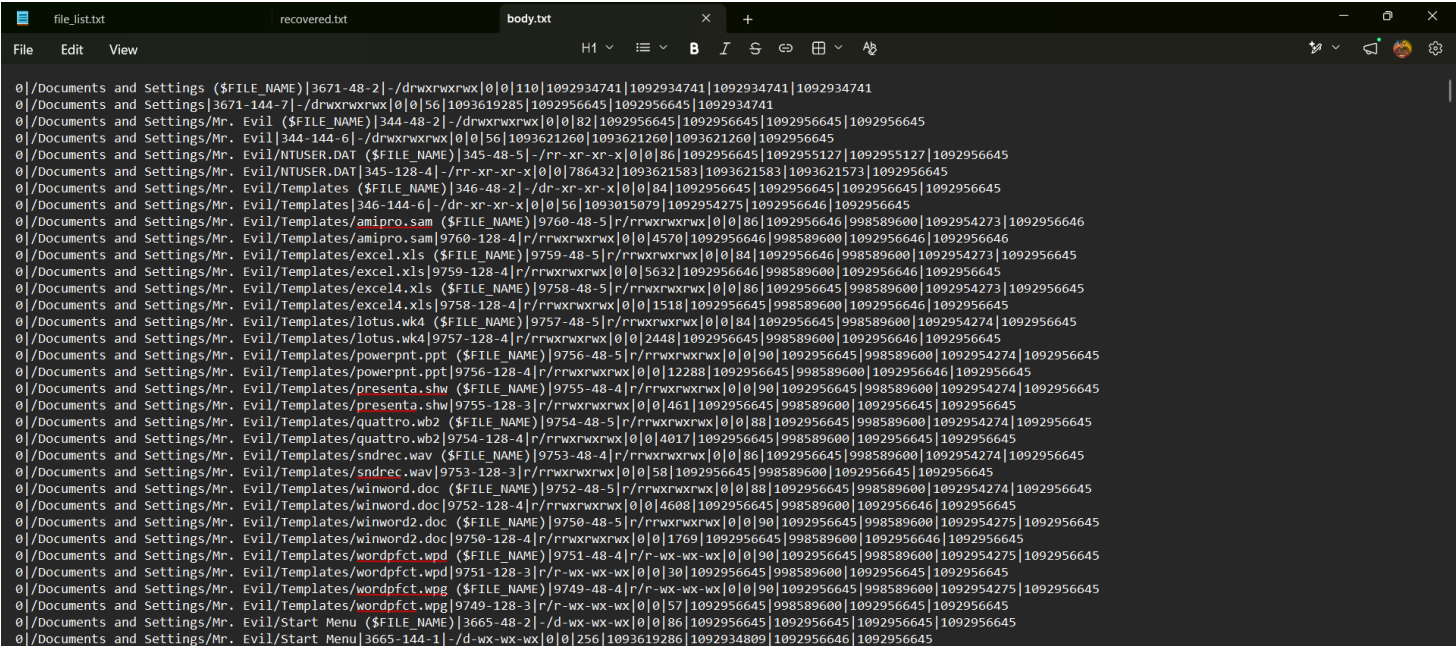
```
C:\Users\kmadh\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>istat -i ewf -o 63 "C:\Users\kmadh\Downloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01" 150
MFT Entry Header Values:
Entry: 150      Sequence: 150
$LogFile Sequence Number: 9239131
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ( )
Version 16 of 16943392
Created: 1999-04-24 03:52:00.000000000 (India Standard Time)
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
MFT Modified: 2004-08-19 22:32:11.098512000 (India Standard Time)
Accessed: 2004-08-18 05:30:00.000000000 (India Standard Time)

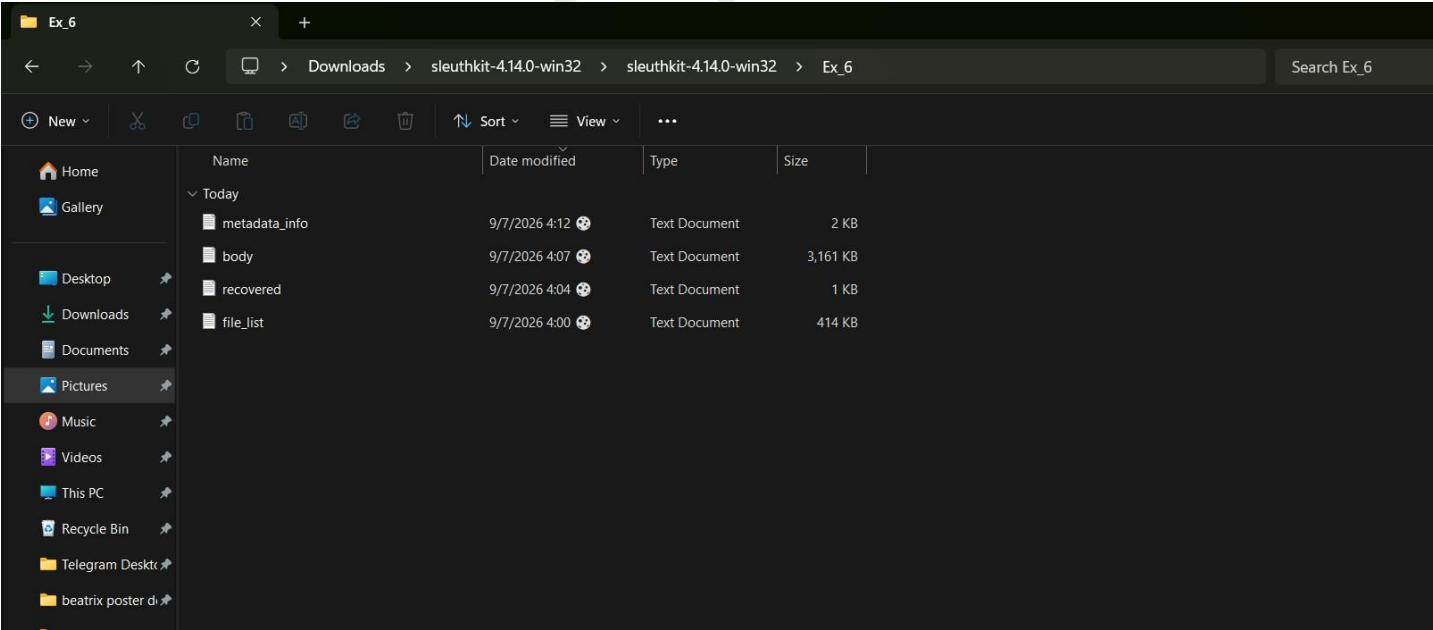
$FILE_NAME Attribute Values:
Flags: Archive
Name: ADOFRE15.REG
Parent MFT Entry: 137 Sequence: 137
Allocated Size: 1024 Actual Size: 640
Created: 1999-04-24 03:52:00.000000000 (India Standard Time)
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
MFT Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
Accessed: 2004-08-18 05:30:00.000000000 (India Standard Time)
```

Generating Timeline:

```
C:\Users\kmadh\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf
-o 63 -m / -r "C:\Users\kmadh\Downloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).
E01" > body.txt
```



Gathering all the generated Files(Report):



Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: Successfully examined file systems, metadata, timelines, and deleted files using The Sleuth Kit (TSK).